

Pan-Cancer Pharmacogenomic Analysis of Patient-Derived Tumor Cells Using Clinically Relevant Drug Exposures



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ABSTRACT

As a result of tumor heterogeneity and solid cancers harboring multiple molecular defects, precision medicine platforms in oncology are most effective when both genetic and pharmacologic determinants of a tumor are evaluated. Expandable patient-derived xenograft (PDX) mouse tumor and corresponding PDX culture (PDXC) models recapitulate many of the biological and genetic characteristics of the original patient tumor, allowing for a comprehensive pharmacogenomic analysis. Here, the somatic mutations of 23 matched patient tumor and PDX samples encompassing four cancers were first evaluated using next-generation sequencing (NGS). 19 antitumor agents were evaluated across 78 patient-derived tumor cultures using clinically relevant drug exposures. A binarization threshold sensitivity classification determined in culture (PDXC) was used to identify tumors that best respond to

drug *in vivo* (PDX). Using this sensitivity classification, logic models of DNA mutations were developed for 19 antitumor agents to predict drug response. We determined that the concordance of somatic mutations across patient and corresponding PDX samples increased as variant allele frequency increased. Notable individual PDXC responses to specific drugs, as well as lineage-specific drug responses were identified. Robust responses identified in PDXC were recapitulated *in vivo* in PDX-bearing mice and logic modeling determined somatic gene mutation(s) defining response to specific antitumor agents. In conclusion, combining NGS of primary patient tumors, high-throughput drug screen using clinically relevant doses, and logic modeling, can provide a platform for understanding response to therapeutic drugs targeting cancer.

Introduction

Targeted therapies matched to the specific molecular profile of a patient's tumor have significantly improved the efficacy of cancer treatments and enabled the cancer care personalized medicine revolution (1, 2). Molecular analyses of archived tumor specimens from patients only reveal a static snapshot of the pathways driving cancer progression and do not model the real-time dynamic interplay between the intact tumor and a therapeutic agent. A combination of next-generation sequencing (NGS) in archived tumor specimens and determining the therapeutic response in matched live patient-derived tumor models may therefore provide a more robust platform for personalizing treatments for patients with advanced cancers (3–8). In the development of effective personalized therapy platforms for patients with cancer, research efforts have focused on the use of patient-derived xenograft (PDX) mouse models, and PDX cultures (PDXC) created from PDX models. In a pan-cancer study, researchers compared drug response of 92 patients with response observed in their corresponding PDX model (9). Their analysis showed that drug

response in PDX models predicted the response in patients 87% of the time, even as patients underwent several cycles of therapy. This is in agreement with other studies demonstrating the utility of PDX to predict the clinical response of the patient (10–13).

The interrogation of multiple drug activities in PDX models is labor intensive and cost prohibitive, and patient-derived tumor cells (PDC) are not expandable resources, whereas PDXC offer a lower cost expandable resource for pharmacogenomic analysis. Through the California Pacific Medical Center Research Institute Cancer Avatar Project, we have developed over 300 PDX and corresponding PDXC models across 9 different types of cancer, many of which have been tested against multiple drugs in high-throughput drug screening (HTDS). The HTDS platform evaluates drug activities against 3-dimensional tumorspheres, and uses clinically relevant drug exposures (14). Pharmacologic data for PDXC models is also accompanied by corresponding DNA mutational profiles using NGS.

In this investigation, we studied the fidelity of cancer-driving genes across patient tumor and PDX encompassing multiple cancers. We also evaluated the concordance of somatic mutations across four cancers, and found that in all cancers, concordance increases as variant allele frequency (VAF) increases. Seventy-eight patient-derived tumor cultures were analyzed for response to 19 antitumor agents. Notable individual PDXC responses to specific drugs, as well as lineage specific drug responses were identified, and robust responses identified in PDXC were recapitulated *in vivo* in PDX-bearing mice.

Finally, we used Logic Optimization for Binary Input to Continuous Output (LOBICO) to combine pharmacologic and genomic data to infer small and easily interpretable logic models that explain drug sensitivity based on DNA mutational status. We calculated the area under the curve (AUC) in the sensitivity–specificity plane to assess

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the predictability in drug sensitivity of cell lines based on mutation status (15). The AUC was robustly calculated for the set of logical models of a given maximum complexity using a nested cross-validation approach. Overall, this study provides a unique pan-cancer pharmacogenomic analysis using expandable primary patient-derived tumor models.

Materials and Methods

Generation of PDXs and PDCs

PDCs

Upon receiving human tumor tissue, samples were mechanically disrupted and cultured as previously described (14, 16). Written informed consent was obtained from all subjects involved in the study, which was conducted according to the guidelines of the Declaration of Helsinki, and approved by the Institutional Review Board of Sutter Health (protocol code 2015.059–1 approved on 03/10/22).

PDX generation and passaging

PDCs and PDXCs were passaged *in vivo* as previously described (14, 16).

HTDS

Development of inhibitor screen

All PDXCs or PDCs were plated in 384-well round bottom microplates (Corning) within 3 to 7 days after processing the original tumor material or thawing frozen cells. They were then allowed to acclimate for 3 days for formation of tumorspheres before addition of drugs. PDXCs were screened as previously described (14) in an HTDS format against a 6-point concentration–response curve of drugs chosen primarily based on the following criteria: 1) The drug is FDA-approved or in clinical trials; 2) The drug is available for research purposes; 3) Pharmacokinetic (PK) data in humans is available. For most drugs, the 6-point concentration–response curve contained drug concentrations beginning with the highest concentration starting with the approximate maximum plasma concentration (C_{max}) reported in published clinical trials (Supplementary Table S1). In a small percentage of drugs in historical screens, where there was significant HTDS data, the concentration was kept as long as it did not vary more than 3-fold of the published C_{max} , the exceptions being vorinostat and gedatolisib which varied by 5- to 10-fold.

Drugs

All compounds tested in this study were obtained through SelleckChem (Houston, TX). The detailed information about the drugs, including the structure, can be found using the publicly available search engine on their homepage as well as in the Supplementary Table S1.

Quantification of drug response

The resulting drug-response data were used to calculate area under the pharmacologic curve (AUC) and corresponding confidence limits with the program Prism (GraphPad) using the formula $\Delta X \cdot (Y_1 + Y_2) / 2$ as previously described (14). When taking into consideration the AUC while testing six drug concentrations, the value is between 0 (complete inhibition of cell viability) and 270 (no effect). A value greater than 270 represents cell growth.

Growth index

A fixed-interval approach (17), was used based either on CellTiter-Glo luminescence (indirect measurement of cell number) at the start (x_0) and end of the experiment (x_{ctrl} at 72 hours) was obtained. Time

Zero plate contains a single column with cells. Time zero is obtained by averaging all luminescence values within 95% confidence interval (CI) after z-score normalization. The growth index was calculated as followed. Obtain average for all controls in a plate that fall within the 95% CI after z score normalization for that plate. Obtain average for all averages across the plates. Obtain growth index by dividing the final average by the time zero value.

Cancer gene panel-based NGS

Genomic DNA was extracted from fresh human and mouse PDX tissue samples and sequenced as previously described (14). A library targeting 212 amplicons in 48 genes was generated using the Illumina TruSeq Amplicon – Cancer Panel that provides pre-designed, optimized oligonucleotide probes for sequencing mutational hotspots in >35 kilobases of target genomic sequence.

NGS

The gene targets covered by the Illumina MiSeq were as follows: *ABL1*, *AKT1*, *ALK*, *adenomatous polyposis coli (APC)*, *ATM*, *BRAF*, *CDH1*, *CDKN2A*, *CSF1R*, *CTNNB1*, *EGFR*, *ERBB2*, *ERBB4*, *FBXW7*, *FGFR1*, *FGFR2*, *FGFR3*, *FLT3*, *GNA11*, *GNAQ*, *GNAS*, *HNFI1A*, *HRAS*, *IDH1*, *JAK2*, *JAK3*, *KDR*, *KIT*, *KRAS*, *MET*, *MLH1*, *MPL*, *NOTCH1*, *NPM1*, *NRAS*, *PDGFRA*, *PIK3CA*, *PTEN*, *PTPN11*, *RBI*, *RET*, *SMAD4*, *SMARCB1*, *SMO*, *SRC*, *STK11*, *TP53*, *VHL*. Concordance between the original sample and its derivative was calculated as followed: Concordance: $C = 100\% \cdot (x / y)$

x = number of variants confirmed in both the initial tumor tissue AND the derivative tissue

y = number of similar variants confirmed in initial tumor tissue OR the derivative tissue

When there is a threshold (0, 10%, 25%) then there are different definitions of when variants are confirmed in both the initial tumor tissue and the derivative mouse tissue. The variants are confirmed in both tissues if the variant is above the threshold in at least one tissue and above zero in the other tissue.

LOBICO

LOBICO is a machine learning approach based on integer linear programming for solving classification problems with weights (15). Applied to the pharmacologic and genomic data set, LOBICO required three main inputs during the training stage: (i) a list of binary gene mutations (on/off) for each cell line (mutation matrix), (ii) a binarization threshold indicating whether a certain cell line is sensitive or resistant to a given drug, and (iii) a list of weights set to $|\log(\text{AUC}) - b|$ for each cell line and drug.

Weighting of gene

The “importance” of a gene for a given drug is calculated as follows:

1. For each drug we have the set F of Pareto-optimal formulae according to the test set (these are the formulae that make up the ROC curve for the test data). These formulae have been calculated with LOBICO.
2. The weight of a gene in a formula is calculated as follows:
Each gene that is not mutated (e.g., $\sim BRAF$ or wild-type *BRAF*) counts as -1 . The individual weights of a gene (mutated or wild-type) that occurs multiple times in a formula are added. For instance, in the formula $BRAF \& \sim KRAS \mid \sim BRAF \& KIT$, *BRAF* has a weight of $1 + (-1) = 0$, while *KRAS* has a weight of -1 and *KIT* has a weight of 1 .

- The importance of a gene for a given drug is calculated as the sum of the weights for that gene over all Pareto-optimal formulae F divided by the number of formulae in F .

Binarization threshold

A cell line is defined as sensitive (resistant) based on its AUC value being lower (larger) than a drug-specific threshold value b . This binarization threshold is calculated (separately for each drug) in four steps (K16). The description below summarizes the main approach adopted in this work and it highlights any significant differences to K16:

Step 1 (Up-sampling): For each of a set of N cell lines, 1,000 AUC values are generated on the basis of the measured AUC value of the cell line and its error assuming a normal distribution in $\log(\text{AUC})$. K16 use AUC instead of $\log(\text{AUC})$.

Step 2 (Density Estimate): Subsequently, a 1-dimensional probability density function (pdf) of $\log(\text{AUC})$ is calculated from the set of $1,000 \times N \log(\text{AUC})$ values via a Gaussian kernel density estimate with a bandwidth of 0.5.

Step 3 (Modeling of resistant cell lines): The $\log(\text{AUC})$ values of the resistant cell lines are modeled as a normal distribution g with its mean μ given by the maximum of the pdf. The median distance from μ of all (upsampled) data points in the interval $[\theta, \mu]$ is adopted as the standard deviation σ of g . Here, θ is an auxiliary parameter used as a proxy for b and calculated as described in K16.

Step 4 (Estimating b): The binarization threshold is set to the 10th percentile of the distribution g . We chose the larger percentile compared with the 5th percentile adopted by K16 to boost the fraction of sensitive cell lines given the smaller number of cell lines (30–71 per drug) of our sample.

LOBICO

Following the training stage, LOBICO provides, for each drug d , a logical formula f_d in disjunctive normal form with its propositional variables given by the binary gene mutations (on/off interpreted as True/False). The formula f_d can be used to predict whether a new cell line is likely sensitive or resistant to drug d . The on/off state of binary gene mutations of the given cell line are inserted into the propositional variables of f_d and subsequently the formula is evaluated resulting in a True or False statement. If f_d evaluates to True (False), then the new cell line is predicted to be sensitive (resistant) to drug d . The specific formula f_d provided by LOBICO is optimal in minimizing the weighted mis-match between the predicted sensitivity and the actual sensitivity of the provided cell lines to drug d . The complexity of the formula f_d is determined by two hyper-parameters K , M . Here, K is the maximum number of conjunctions in the disjunction, while M is the maximum number of literals per conjunction.

Figure of merit - AUC

LOBICO allows the addition of constraints on the minimal sensitivity and minimal specificity of the logical formulae. Following K16, we vary the lower bounds on the sensitivity and specificity in steps of 0.05 to find formulae with the best possible values of sensitivity, specificity, or both (i.e., Pareto efficient formulae) for any given drug with the help of a nested cross-validation. The sensitivity and specificity values of these formulae trace out a curve in sensitivity-specificity space and the AUC is used as figure of merit. During the nested cross-validation, an inner 4-fold cross-validation is employed to select the K , M hyper-parameters that result in the highest average (over the inner folds) AUC. Given the tuned hyper-parameters, LOBICO is then trained during the outer 5-fold cross-validation to provide a formula for each sensitivity and specificity bound (provided a

formula satisfying the constraints can be found). Pareto efficient formulae are identified among these formulae by solving the maximum vector problem. A formula f is Pareto efficient if there is no other formula f' that is better than f in the following sense. A formula f is better than a formula f' if: (i) f has higher sensitivity than f' and its specificity is at least as high as the one of f' , (ii) f has higher specificity than f' and its sensitivity is at least as high as the one of f' , or (iii) f has the same sensitivity and specificity as f' but contains a smaller number of literals than f' . Subsequently, the sensitivities and specificities of the set of Pareto efficient formulae is calculated for each fold (test set) of the outer cross-validation. The curve traced by the sensitivity and specificity values of the Pareto efficient formulae is the Pareto frontier. Finally, the figure of merit and its error are given by the mean and by the standard deviation of the area under the Pareto frontier for the 5 folds of the outer cross-validation.

Code availability

The scripts used to carry out the threshold and LOBICO analysis can be found at:

<https://github.com/sch2007/pancancer-pharmacogenomic-analysis.git>.

LOBICO (Knijnenburg and colleagues 2016) is available at <https://github.com/tknijnen/LOBICO>.

In vivo treatment studies

In vivo studies were carried out in accordance with the NIH guidelines and a California Pacific Medical Center (CPMC)-approved Institutional Animal Care and Use Committee protocol (#18.08.03). Four to five, 8- to 10-week-old female NSG mice (Jackson Laboratories, Sacramento, CA) bearing subcutaneous PDX tumors were randomized according to initial tumor volume. They were treated with either vehicle or 25 mg/kg ganetespib. Drugs were diluted in 4% DMSO, 4% Tween-80, and 92% saline. Treatment began when PDX tumor volumes reached approximately 100 mm³. PDX-bearing NSG mice were treated intraperitoneally daily for 5 consecutive days for approximately 4 weeks or until PDX tumor volumes reached 2,000 mm³. Tumors were measured approximately once to twice weekly by an operator blinded to the treatment groups. Mice were housed in a single ventilated cage in groups of four, provided environmental enrichment materials, and free access to water and food. At the completion of *in vivo* experiments, all animals were humanely killed using CO₂ overdose followed by thoracotomy as outline by the American Veterinary Medical Association guidelines for the euthanasia of animals.

Tumor volume was calculated as $(\text{length} \times \text{width}^2)/2$. The response was determined by comparing tumor volume change at time t to its baseline: % tumor volume change = $\Delta \text{Vol}_t = 100\% \times ((V_t - V_{\text{initial}})/V_{\text{initial}})$. For comparison of drug response of vemurafenib and cobimetinib across PDX, the volume of the treated group for the last measurement day was compared with the volume of the vehicle (control) group and tumor volume (% Control) was determined: $100\% \times [1 + (\text{Tumor volume}_{\text{treated}} - \text{Tumor volume}_{\text{vehicle}})/\text{Tumor volume}_{\text{vehicle}}]$. Four to 5 mice were used for studies based on sample size calculations from preliminary studies.

Statistical analyses

For *in vivo* experiments, differences in tumor growth between treatment groups were evaluated using Two-way ANOVA repeated measures, and Tukey multiple comparisons test.

Data availability

The data generated in this study are available within the article and its Supplementary Data files.

Results

Establishment of PDXC models from tumors

78 patient-derived tumor cultures were established across nine cancers: glioblastoma, colorectal, ovarian, pancreatic, melanoma, cholangiocarcinoma, gall bladder, and breast (Fig. 1A). A majority of the cultures were PDXC developed from PDX, but in some cases enough tissue was collected to develop primary patient-derived tumor cultures (PDC) for analysis. As previously demonstrated, for specific drugs there is highly concordant drug sensitivity across derivatives (3). For generation of PDX, human tumor tissue samples were received within 1 to 2 hours after resection and tumor material was implanted in NSG mice for expansion (Fig. 1B). After successful expansion, PDX tissue was collected from mice and PDXC were established (Fig. 1B). PDX and corresponding PDXC were analyzed for somatic mutation using the Illumina TruSeq Amplicon 48 gene cancer panel (Fig. 1C), and PDXC were evaluated for sensitivity to a panel of drugs using HTDS (Fig. 1D). The pharmacologic and genomic data are combined to create logic models to predict sensitivity of a given cell line to a drug based on DNA mutation status (Fig. 1E).

The 78 tumors were obtained from 34 men and 44 women (Supplementary Table S2), with ages ranging from 40 to 90s, and obtained from various sites. Patients were treatment naive or had specific treatment histories for each cancer. The tumor samples collected, which were used to derive PDC, PDX, and PDXC, reflect patient populations receiving current standard of care.

Concordance of somatic mutations across patients and PDXs correlates with VAF

For four cancers, melanoma (Fig. 2A), glioblastoma (Fig. 2B), colorectal (Fig. 2C), and pancreatic (Fig. 2D), there was enough tumor tissue available from the original patient sample and corresponding PDX models to perform NGS. A total of 23 matched samples were analyzed. The nomenclature used for xenograft passaged tumors was X (first generation) and X1 (second generation), corresponding to serial passaging *in vivo*. A sequencing library targeting 212 amplicons in 48 genes was generated using the Illumina TruSeq Amplicon – Cancer Panel. In agreement with previously published Cancer Genome analyses (18–21) lineage specific mutations were identified for individual cancers. For example, tumor tissues from the patients and their corresponding PDXs in melanoma contained *BRAF*, *NRAS*, *PTEN*, *KIT*, and *TP53* mutations, GBM samples contained *PTEN* and *EGFR* mutations, colorectal cancer samples contained *APC* and *EGFR* mutations, and all pancreatic cancer samples contained *KRAS* mutations. Among the five ovarian cancer specimens tested in this study, three were high-grade serous ovarian cancer (HGSOC), one was later categorized as squamous cell carcinoma of the cervix (OVCA0033) and one was a yolk sac tumor (OVCA0027). The HGSOC commonly carry *TP53* mutations, however, these mutations were not detected in our samples, which may be the result of using a cancer panel that does not cover all exons (14).

We previously published in melanoma that an increase in VAF corresponded to an increase in stability of specific mutations across the patient tumor, corresponding PDX, and subsequent generations of

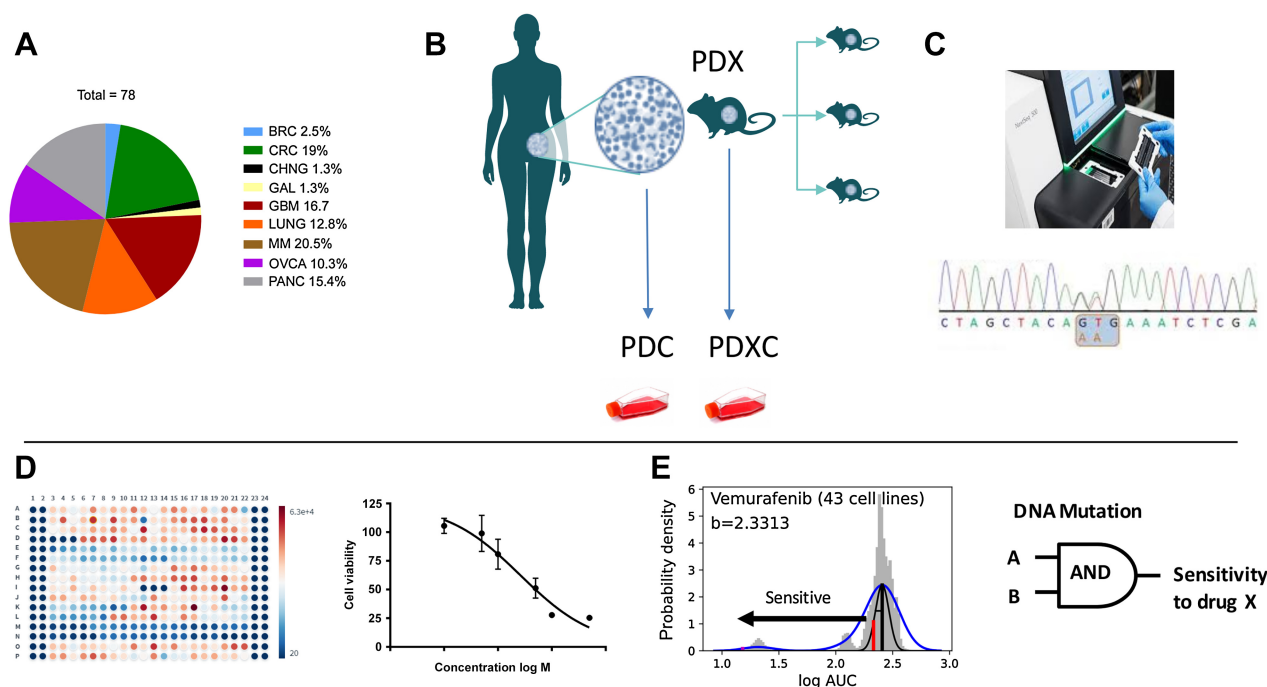


Figure 1.

Pharmacogenomic platform for analysis of patient-derived tumor cultures. **A**, Tumor tissue across nine cancers was collected from patients: glioblastoma (GBM), colorectal (CRC), ovarian (OVCA), pancreatic (PANC), melanoma (MM), cholangiocarcinoma (CHNG), gall bladder (GAL), and breast (BRC). The distribution of cancer types is shown. **B**, The tumors were used to establish PDC and passaged in mice for expansion, and to establish PDXC for a total of 78 unique derivatives. **C**, Patient tumors and derivatives were evaluated for DNA mutations using NGS. **D**, The 78 tumor cultures underwent HTDS (left) to evaluate sensitivity to 19 antitumor agents. The HTDS heat map of a 384-well plate represents no cell death (dark blue) to complete loss of cell viability (dark red). The right panel represents a typical concentration–response curve generated for a single drug. **E**, Binarization thresholds determined drug sensitive and resistant populations and logic models of DNA mutations were developed to predict drug sensitivity.

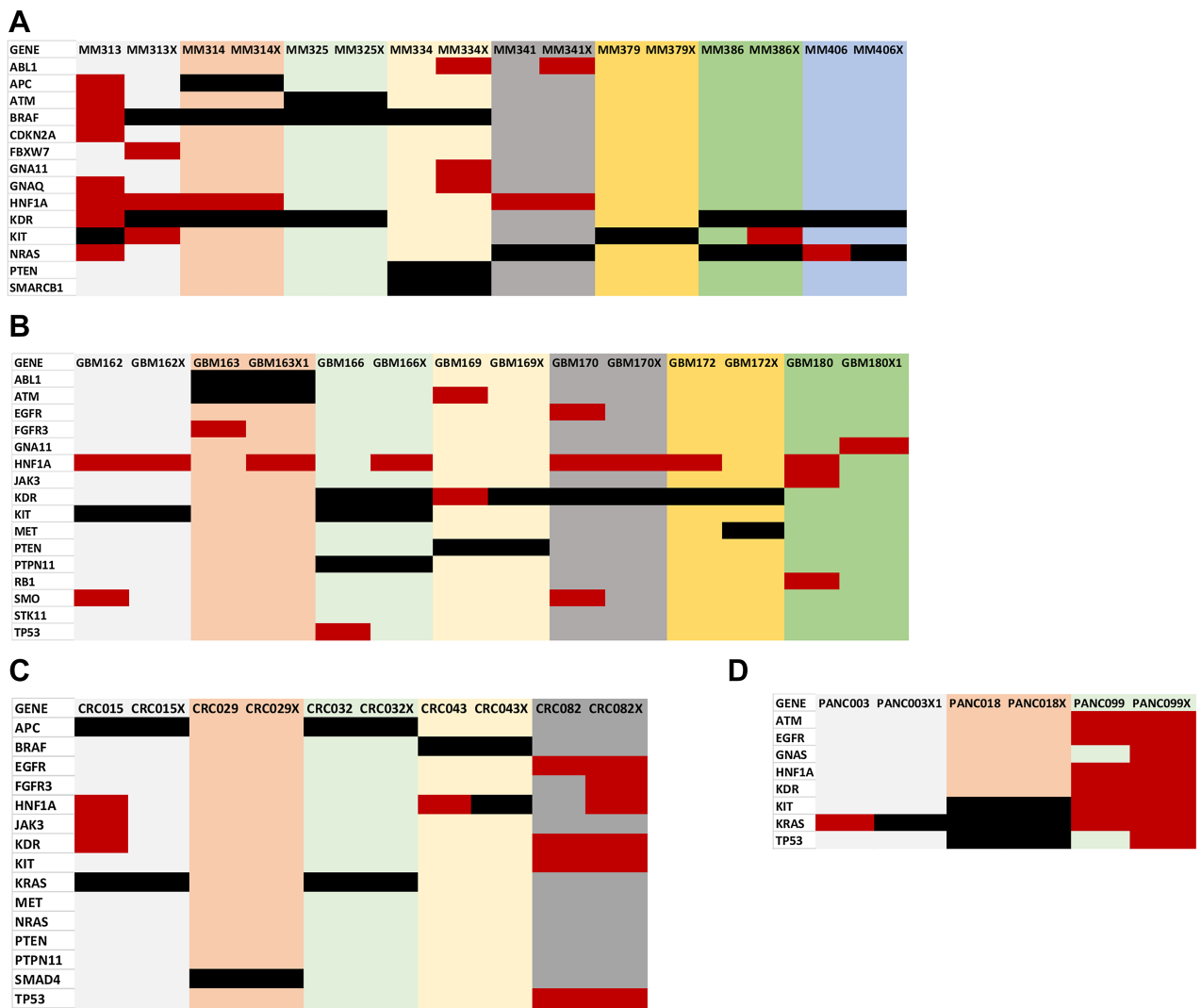


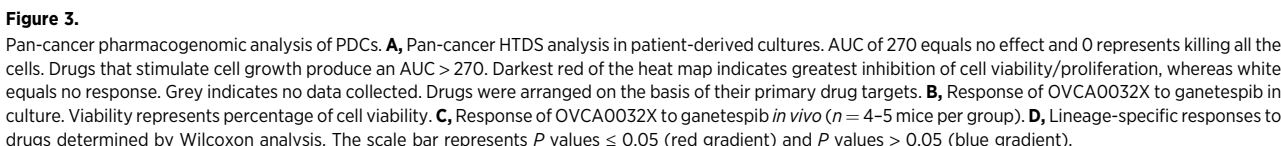
Figure 2. SNV and insertion/deletion variants of DNA mutational hotspots from patient tumor and first or second generation PDX derivatives. Patient tumor and corresponding first-generation (X) PDX or second-generation (X1) PDX from (A) melanoma (B) glioblastoma (C) colorectal, and (D) pancreatic cancer were compared. The matched pairs are grouped using colored columns, for example, MM313 with MM313X (light grey). The mutations identified are listed by row (far left column). Red indicates an allele frequency (AF) 5%–24%. Black indicates an AF ≥ 25%.

PDXs (14). We, therefore, examined this phenomenon across additional cancers including additional melanoma samples. For melanoma, in patient tumor and first generation PDX the total concordance was 58% for VAF ≥ 10% and 91% for VAF ≥ 25%. For glioblastoma, in patient tumor and first generation PDX the total concordance was 65% for VAF ≥ 10% and 100% for VAF ≥ 25%. For colorectal cancer, in patient tumor and first generation PDX the total concordance was 85% for VAF ≥ 10% and 100% for VAF ≥ 25%. For pancreatic cancer, in patient tumor and first generation PDX the total concordance was 100% for VAF ≥ 10% and 100% for VAF ≥ 25%. Our data demonstrate that PDXs harbor somatic mutations in cancer-driving genes as expected in accordance with TCGA data. Furthermore, the concordance of somatic mutations across patient tumor and matched PDXs increases as VAF increases. The fidelity of cancer-driving gene mutations across human and PDX samples suggests that responses to antitumor agents across derivatives would be conserved.

Evaluation of PDXC response to antitumor agents using HTDS

19 antitumor agents were evaluated across 78 patient-derived tumor cultures, constituting 9 different cancers (Fig. 3A). Tumors were cultured as tumorspheres in 384-well plates, treated with drugs for 3 days and assessed for cell viability. Because a precision medicine platform was being developed, we chose our maximum concentration in the assay to correspond to the maximum plasma concentration (C_{max}) at the highest single dose recommended on the basis of clinical trials when the information was available, and when the drug was not limited by solubility (Supplementary Table S1). After drug treatment, the area under the dose-response curve (AUC) with corresponding CIs was calculated. As cell viability/proliferation is decreased, the AUC is reduced (22, 23).

Past studies, and recent large scale screening efforts, demonstrate that AUC is more effective than either calculating the potency and efficacy of a drug across multiple cancer lines, because AUC takes into



epithelial carcinoma with a wild-type p53. In PDXC OVCA0032X, ganetespib produced a potent and efficacious inhibition of cell viability/proliferation. In agreement with the activity observed in PDXC, we found that ganetespib produced a sustained inhibition of tumor growth in mice bearing PDX OVCA0032X (Fig. 3C).

It has been reported in serum-derived cell lines that for specific drugs the growth rate (GR) of cells may confound the comparison of standard pharmacology measures (IC_{50} , Emax, AUC) between cell lines (17). Modified pharmacologic analysis (e.g., GR_{50} and GR_{max}) to account for proliferation rate were proposed and were expected to more correctly facilitate the discovery of biomarkers that predict drug

sensitivity. As opposed to serum-derived cancer cell lines that undergo at least one cell division over a period of 3 days, 25% ($n = 40$) of PDXC show limited cell division as assessed by calculating the growth index (Supplementary Fig. S1). These growth characteristics are more representative of patient tumors that are often slow growing. GR analysis was therefore not appropriate for many PDXC, because the underlying equations are dependent on a significant increase in the cell population in the control groups over time (17). We therefore calculated the AUC and standard pharmacologic measures to compare drug sensitivity across all primary patient-derived tumor models tested.

Binarization threshold determination to classify drug sensitivity and resistance for logic modeling

We used LOBICO, a computational approach that infers small and easily interpretable logic models of binary input features that explain a continuous output variable (15). In the context of this investigation, this model was used to uncover logic combinations of multiple DNA

mutations that would be more predictive of drug response than single DNA mutation predictors. Single predictors (e.g., a DNA mutation) are generally less accurate predicting drug response because of the complexity of biological pathways driving cancer.

Our analysis was based on screening 19 drugs across 78 patient-derived tumor cultures from 9 different cancers, and each of the tumors underwent a targeted resequencing assay for detection of somatic mutations in cancer genome hotspots. The AUC with corresponding confidence limits was obtained after 3 days treatment with a drug and served as the continuous output variable. LOBICO was applied to the AUCs and corresponding confidence limits of each drug separately. For the first step of LOBICO, a binarization threshold for the output variable is calculated (Fig. 4A; Supplementary Fig. S2, Supplementary Table S3). This threshold is used to separate the PDXC into two groups: PDXC sensitive and PDXC resistant to drug (Fig. 4B; Supplementary Table S4). LOBICO finds an optimal logic function of binary mutation features that minimizes the error, which is defined as

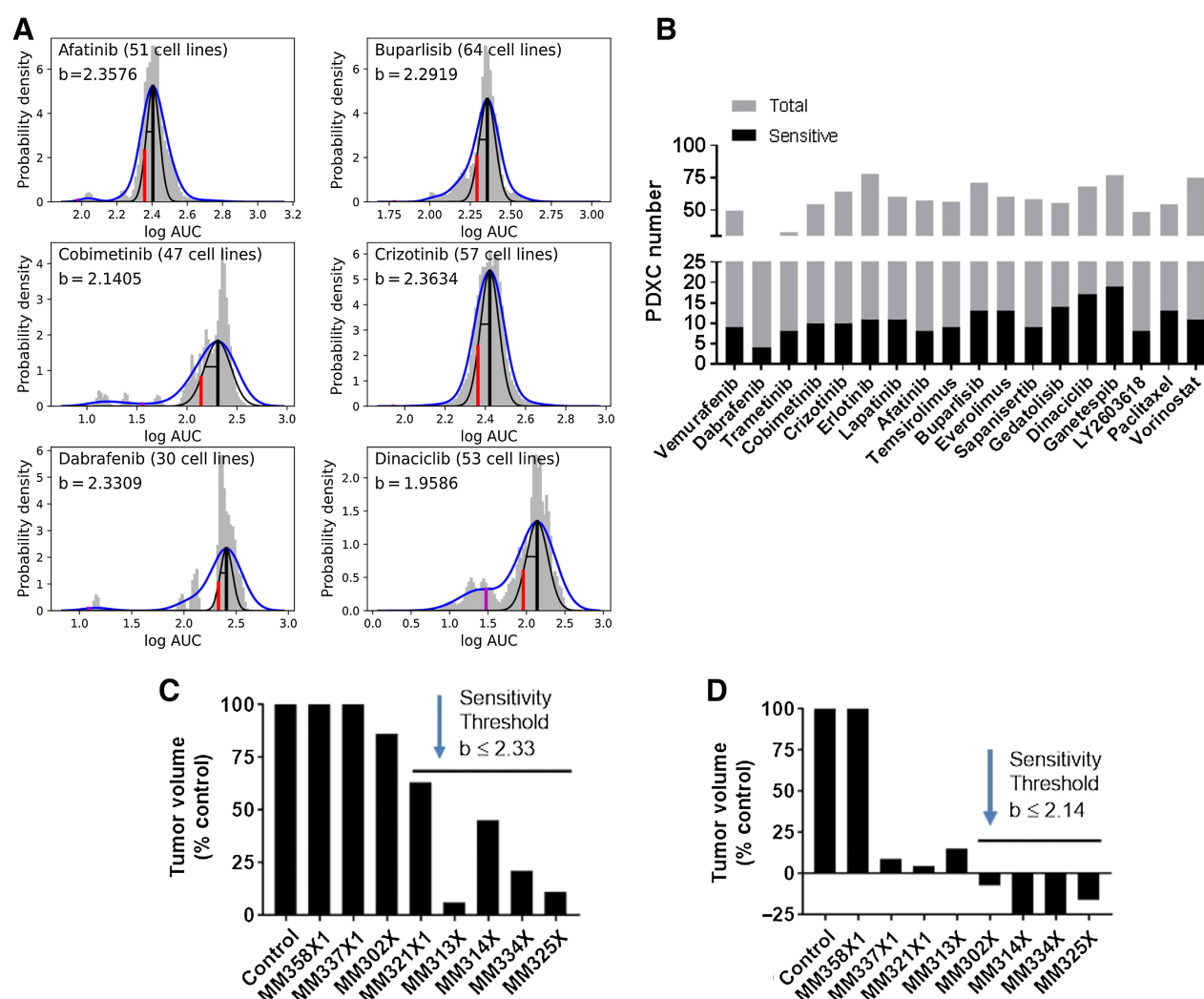


Figure 4.

Binarization threshold determination of drug sensitivity. **A**, Bar histogram of upscaled AUC values (grey) and kernel density estimate (blue). Binarization threshold b (red) marks the divide between the population of sensitive (smaller AUC values) and resistant (larger AUC values) cancer cells. The x-axis shows $\log(\text{AUC})$ values for cell lines for a particular drug, and the y-axis shows the probability density. **B**, Percentage of sensitive and resistant patient-derived cancer cells tested for each drug. Response of 8 melanoma PDX to **(C)** vemurafenib and **(D)** cobimetinib. The sensitivity threshold denotes tumor models derived from PDXC that met the criteria of "sensitive" to drug.

the sum of the weighted misclassified cell lines (15). Importantly, while this method has been used to define resistant and sensitive populations of cell lines, it has never been validated *in vivo* to determine the extent to which the classification predicts response of a tumor in a preclinical mouse xenograft model. Validating the drug resistant and sensitive classification obtained using PDXC (culture), in PDX (*in vivo*) models, provides increased confidence in the relevance of the binarization threshold findings obtained in culture (PDXC).

We chose PDXC and corresponding PDX derived from 8 *BRAF*-mutant melanoma for our analysis. We evaluated the resistant and sensitive classification of vemurafenib and cobimetinib in PDXC to the response observed *in vivo* (Fig. 4C and D). The AUC-derived binarization threshold predicted for vemurafenib and cobimetinib in the PDXC was 2.33 and 2.14, respectively. The log AUC values were: vemurafenib (MM358X1 = 2.34, MM337X2 = 2.42, MM321X1 = 2.22, MM302X = 2.39, MM313X = 2.11, MM314X1 = 2.0, MM334X = 1.34, MM325X = 1.29) and cobimetinib (MM358X1 = 2.32, MM337X1 = 2.15, MM321X1 = 2.22, MM302X = 1.79, MM313X = 2.13, MM314X1 = 1.71, MM334X = 1.39, MM325X = 1.12). The log AUC values for MM358X1, MM337X2, MM321X1, and MM302 were derived previously (14).

We then compared the response of the corresponding PDX models to vemurafenib and cobimetinib using previously published data from our group (14). The drug responses *in vivo* were originally evaluated by measuring tumor volume (mm³). In the current study, the tumor volume (mm³) of the treated group for the last measurement day was compared with the tumor volume of the vehicle (control) group and tumor volume (% Control) was calculated. These data were used to cross compare the response of all 8 PDX to vemurafenib and cobimetinib (Fig. 4C and D). Cobimetinib produced robust inhibition of tumor growth in the models falling under the sensitive classification (Fig. 4D). Inhibition of tumor progression was also observed in a subset of models falling under the insensitive classification, but to a lesser extent than observed in the sensitive classification. For vemurafenib, the classifications defined tumors sensitive and resistant to treatment *in vivo* (Fig. 4C). Overall, the binarization threshold sensitive classification determined in culture (PDXC) identifies tumors that respond to drug *in vivo* (PDX).

Logic models of DNA mutations to predict drug sensitivity

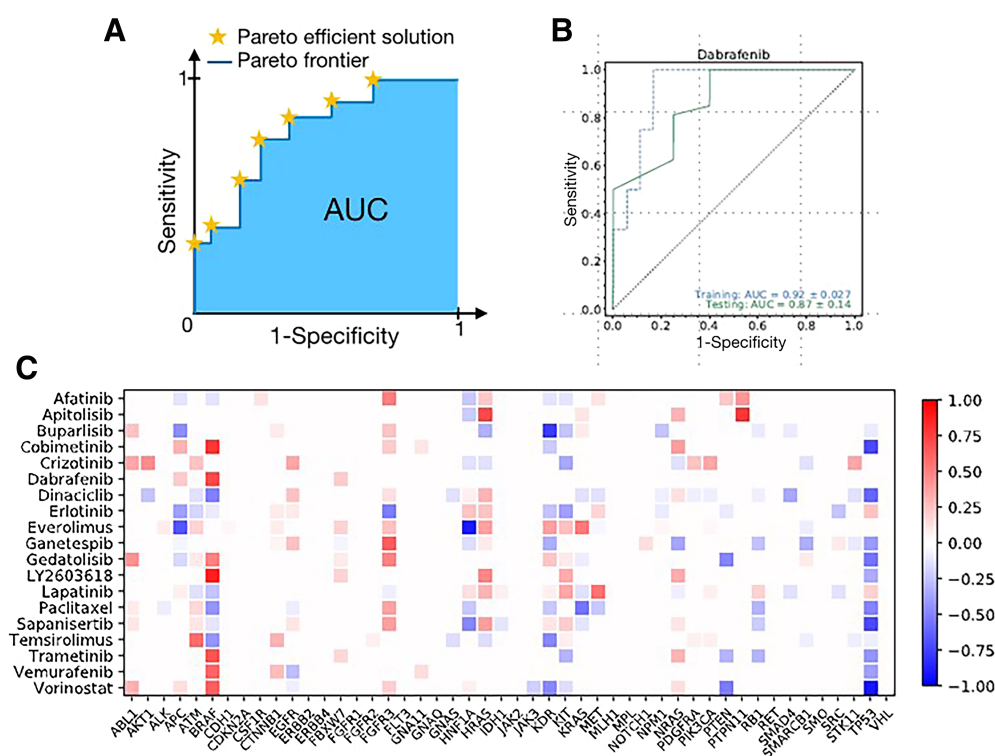
Once binarization threshold sensitive classification were determined, we next used LOBICO to address the question of how well drug sensitivity can be explained by logic models of genes evaluated in the cancer gene panel-based NGS. The mutational data for each sample is inputted as a binary feature, where 0 is the absence of a DNA mutation (e.g., wild-type *BRAF*) and 1 is the presence of a DNA mutation (e.g., *BRAF* V600E) (Supplementary Table S5). During training, which is carried out separately for each drug, LOBICO requires both the mutational data and the AUC data for the cell lines. The AUC data are provided as raw data calculated using a standard formula (Materials and Methods) or as scaled data from 0 to 1 (Supplementary Table S5). Our analysis incorporated the raw AUC data. Following the training stage, LOBICO provides, for each drug *d*, logical formulae in disjunctive normal form with its propositional variables given by the binary gene mutations (on/off interpreted as True/False). These formulae predict whether a new cell line is likely sensitive or resistant to the considered drug depending on whether they evaluate to True or False. The logic models provided by LOBICO are optimal in minimizing the weighted mis-match between the predicted sensitivity and the actual sensitivity of the provided cell lines to drug *d*.

LOBICO allows to put constraints on the sensitivity and specificity of logic models. Following (15), we vary the lower bounds on the sensitivity and specificity in steps of 0.05 to find logic models with the best possible values of sensitivity, specificity, or both for any given drug. The sensitivity and specificity values of these models trace out a curve in sensitivity–specificity space, the Pareto frontier, and the AUC is then interpreted as figure of merit (Fig. 5A and B; Supplementary Fig. S3). To minimize overfitting and arrive at a more robust estimate of the figure of merit we employ nested cross-validation, consisting of an inner 4-fold cross-validation to tune the hyper-parameters of the logic models and an outer 5-fold cross-validation to robustly estimate the figure of merit and its error (Supplementary Table S6).

Figure 5B shows the Pareto frontier for the drug dabrafenib. The Pareto frontier for the additional drugs is presented in Supplementary Fig. S3. The area under the specificity – sensitivity curve for dabrafenib is 0.92 (±0.03) if applied to the cell lines in the training and validation set, and 0.87 (±0.14) if applied to the cell lines in the test set during cross-validation. This indicates that logical models are able to predict whether cell lines are resistant or not to those drugs with reasonable accuracy. One of the high scoring logic models for dabrafenib is the single gene *BRAF* as is expected for this selective inhibitor, validating the prediction potential of the model. The predictions for all drugs are presented in Supplementary Table S6.

The curve in sensitivity–specificity space (Pareto frontier) shown in Fig. 5B is associated with a list F of formulae with high sensitivity, specificity or both. By analyzing these formulae for each drug, we can identify genes that help in identifying sensitive and resistant cell lines. The importance of a gene for a given drug is calculated as follows. First, the weight of a gene is calculated for each formula *f* from F. Each gene is assigned a weight, where each mutated gene (e.g., *BRAF* V600E) counts as 1. A gene such as *BRAF* could have various mutations based on the NGS panel, which covers multiple mutation hotspots (e.g., *BRAF* V600E, *BRAF* V600K, etc.). Each gene that is not mutated (e.g., *~BRAF* or wild-type *BRAF*) counts -1 . If a gene occurs multiple times in a formula the individual weights are added (an alternative option would be to average the weights). For instance, in the formula *BRAF* & *~KRAS* | *~BRAF* & *KIT*, *BRAF* has weight 1 and *~BRAF* has a weight of -1 , therefore the sum of *BRAF* = $(1 + -1) = 0$, *~KRAS* has weight -1 , and *KIT* has weight 1. Next, the importance of a gene for a given drug is calculated as the sum of the weights for that gene for each formula in F divided by the number of formulae in F. Fig. 5C and Supplementary Table S7 show the gene(s) of importance for each drug.

As shown above, cell lines with *BRAF* gene mutations are predicted to be sensitive to treatment with MEK and *BRAF* inhibitors. As expected *BRAF* and MEK inhibitors ranked with high scores for the *BRAF* gene, logic models contained *BRAF* alone or in combination with additional genes and resistance was predicted by *RAS* mutations (ref. 25; See also Supplementary Table S7). *EGFR* was scored for erlotinib but not afatinib, the lack of association was the result of tumors not containing *EGFR* mutations shown to infer sensitization to the drug such as Del19/L858R (26). Of note, a gene of importance for the HSP90 inhibitor ganetespib was *FGFR3*, and many logic models included the presence of *FGFR3* mutations (Supplementary Table S7). Upon review of the individual DNA mutations, we found multiple cancers (pancreatic, lung, and melanoma) defined as sensitive to ganetespib that contained *FGFR3* 3'-prime UTR variants (Supplementary Table S8). *FGFR3* has been shown to be a strong HSP90 client compared with other FGFR receptors, and depends on HSP90 for stability and function (27). In addition, vorinostat ranked with a high score for *BRAF*, and had multiple logic models that included *BRAF* mutations (Supplementary Table S7), thereby demonstrating a lineage-specific

**Figure 5.**

Logic models of drug sensitivity prediction based on DNA mutation status. **A**, Schematic illustrating the calculation of the AUC in the sensitivity – specificity plane. **B**, Dabrafenib sensitivity – specificity curves. **C**, Gene importance as predicted by the LOBICO analysis. Gene importance is defined as the sum of signed occurrences of a gene in a Pareto-optimal formula divided by the number of Pareto-optimal formulae for a given drug. Negated and non-negated genes have a signed occurrence of -1 and $+1$, respectively. For instance, the signed occurrence of the gene *BRAF* in the formula *BRAF* & $\sim$ *KRAS* | $\sim$ *BRAF* & *KIT*, is zero because $1 + (-1) = 0$.

sensitivity in melanoma (Fig. 3D). Vorinostat has emerged as potential single or drug combination therapy in targeting BRAFi-resistant melanoma through modulation of reactive oxygen species (ROS), or production of excessive DNA damage (28, 29). The CHK1 inhibitor LY2603618 also ranked with a high score for *BRAF*, and had multiple logic models that included *BRAF* and *RAS* mutations; the latter finding suggests *RAS*-activating mutations may not limit the efficacy of a CHK1 inhibitor. Indeed, it has been shown that a potent and specific inhibitor of CHK1 is effective in reducing cell viability and colony formation of vemurafenib-resistant melanoma cells in culture and *in vivo* (30). Resistance to multiple PI3K and mTOR inhibitors was observed with the presence of *APC* gene mutations; this is in agreement with past studies of drug sensitivities in PDC (3). Interestingly, the logic models predicting response to the CDK1/2/5/9 inhibitor dinaciclib (31) had the highest sensitivity and specificity with the absence of multiple mutation, for example the absence of mutation in *BRAF*, *AKT*, *KRAS*, and *TP53*.

Discussion

The evaluation of pharmacologic sensitivities of cancer cells paired with their genomic profile provides a powerful platform for precision medicine oncology. Over the last decade, multiple investigations have demonstrated the utility of pharmacogenomics studies in determining gene(s) predicting sensitivity and resistance to single and drug combination therapies (4, 8, 32–35). Many of the large-scale studies have used data from serum-derived cancer cell

lines, including the notable investigations using data derived from CCLE and GDSC studies (8, 32–36). More recent investigations have also used similar approaches using patient-derived primary tumor cultures (3). As opposed to serum-derived cell lines, primary patient-derived tumor cultures have been shown to more faithfully recapitulate the genotype and phenotype of the original patient's tumor (9–13, 37–39). In addition, a recent study showed that a pan-cancer analysis, using over 400 unique PDCs, not propagated in mice, could be used to predict clinical response (3).

Many different statistical and bioinformatics approaches have been examined to enhance pharmacogenomics prediction (3, 40–42). For instance, machine learning based modeling approaches such as logical models, elastic net regression, and random forests as well as ANOVA models have been successfully applied to predict drug response based on genetic information (3, 15, 32). LOBICO has been used to explore pharmacogenomic relationships using serum-derived cell lines (15, 32), but this study represents the first to use LOBICO for analysis of primary patient-derived tumors.

In this investigation, we performed a pharmacogenomic analysis of patient-derived tumor cultures, including PDXC. In contrast to PDC, they are easily expandable for future mechanistic studies both in culture and *in vivo*. In addition, in comparison with studying multiple drug–gene interactions in PDX models, PDXC offer a lower cost resource for this type of analysis and are amenable to HTDS analysis of single and drug combination therapies. We established 78 patient-derived tumor cultures across 9 cancers. Our data demonstrate that the tumors harbor somatic mutations in cancer-driving genes, and the

concordance of somatic mutations across patient tumor and matched PDXs increases as VAF increases. Nineteen antitumor agents were evaluated across 78 patient-derived tumor cultures, constituting 9 different cancers using clinically relevant drug exposures (42). Because the patient-derived tumor cultures do not contain components of the microenvironment, we focused our analysis primarily on agents known to produce antitumor activity by directly targeting tumor cells. Response to PDXC were recapitulated in PDX models. Lineage-specific drug responses were identified, for instance, response of melanoma to BRAF and MEK inhibitors. Additional lineage specific drug responses identified included, vorinostat (HDACi) sensitivity in melanoma, cobimetinib (MEKi) sensitivity in lung cancer, buparlisib (PI3K) and dinaciclib (CDK1/2/5/9) sensitivity in ovarian cancer, and everolimus (mTOR) sensitivity in pancreatic cancer.

We next used logic modeling and a gene score calculation to uncover gene alterations, both single and combination, that predict response to specific drugs. As a first step, binarization thresholds were determined to classify drug sensitivity and resistance in culture for logic modeling. While this sensitivity classification in culture, and additional drug sensitivity classifications based on z-score calculations (3), would theoretically define responses *in vivo*, they have never been tested in detail in preclinical mouse models of cancer. A unique aspect of this study was to evaluate the correlation between the sensitivity classification of BRAF and MEK inhibitors in culture in comparison with response *in vivo*. Overall, the binarization threshold sensitive classification determined in PDXC predicted a drug response in the corresponding PDX model. However, for the MEK inhibitor cobimetinib, some tumors falling under the resistant classification did respond to drug *in vivo*. This classification may be further improved by including more cell lines.

Once binarization threshold sensitive classifications were determined, logic models of DNA mutations were developed to predict drug sensitivity and specificity. The figure of merit is the AUC of the ROC curve in the sensitivity–specificity plane (15). This ROC curve is the Pareto frontier of LOBICO-optimal solutions subject to minimal sensitivity and specificity constraints. Genes of importance for each drug were then determined by calculating the average signed occurrence of that gene (+1 for mutated type, –1 for wild-type) in the LOBICO-optimal formulae.

As expected, BRAF and MEK inhibitors ranked with high scores for the BRAF gene, logic models contained BRAF alone or in combination with additional genes and resistance was predicted by RAS mutations (25). There were multiple novel gene–drug associations, and logic models of interest uncovered by the analysis. For example, a gene of importance for tumor cell sensitivity to ganetespib was FGFR3, and many logic models included the presence of FGFR3 mutations. In FGFR3-driven bladder cancer, ganetespib produces potent cytotoxicity comparable with the pan-FGFR tyrosine kinase inhibitor BGJ398 (43). How the 3′ prime UTR variants observed in our PDXC defined as sensitive to ganetespib affect the HSP90/receptor interaction remains to be determined, but provides an example of how pharmacogenomic analysis using the described platform may identify biomarkers of response to specific drugs. In addition, a gene of importance for sensitivity to vorinostat was BRAF. Multiple logic models for this drug included BRAF mutations, and there was also a lineage specific drug sensitivity in BRAF-wild-type melanoma. This inhibitory activity of vorinostat alone and in combination with other antitumor agents has been shown in a HTDS of melanoma PDC (5); however, there were no genomic correlates described for this activity. As discussed, sensitivity to vorinostat has been reported in melanoma through modulation of ROS, or

production of excessive DNA damage (28, 29). Of interest, BRAF mutation (common in melanoma) was labeled a high predictor of sensitivity to vorinostat, whereas TP53 mutation denoted resistance to the drug. The CHK inhibitor LY2603618 also ranked with a high score for BRAF and this drug has been shown to be effective at inhibiting tumor growth in vemurafenib-resistant melanoma (30). Resistance to PI3K and mTOR inhibitors was also predicted by APC mutations in agreement with past pharmacogenomics analysis of PDC (3). The CDK inhibitor dinaciclib can control the expression of multiple short-lived proteins involved in cancer cell survival (31). It had unique logic models where sensitivity was predicted by the absence of mutations in multiple cancer driving genes including ATM, TP53, BRAF, SMAD4, or was predicted by the presence of EGFR, MET, NRAS, FGFR3 mutations. This may relate to the more global transcriptional inhibition produced by dinaciclib. Intriguingly, our data indicate that many of our ovarian PDXC fall within these classifications leading to lineage-specific sensitivity to dinaciclib.

The limited number of drug-sensitive PDXCs poses a challenge for the present analysis. However, even with this issue, we were able to observe strong drug–gene association with classical genes such as BRAF. In addition, multiple novel gene–drug associations and logic models of interest were uncovered by the analysis. One challenge is that the calculation of the AUC requires at least one sensitive and one resistant cell line per training/validation/testing data set to calculate sensitivities and specificities. As a form of mitigation, we distributed sensitive and resistant cell lines in approximately equal proportions among the training/validation/testing data sets during cross validation. As more PDXC drug–gene data is added to the model, error in AUC estimates will be reduced. Furthermore, increasing the number of cell lines analyzed will also enable the use of more flexible LOBICO formulae (i.e., larger values for K and M) thus potentially increasing the AUC figure of merit. Even with these caveats, multiple lineage and gene (s) specific drug sensitivities were identified as described above, including novel associations.

Overall, we performed a pharmacologic analysis of antitumor agents used at clinically relevant doses in expandable primary patient-derived tumors across cancers. Using logic modeling, the pharmacologic data were combined with DNA mutational analysis of cancer driving genes in the primary patient-derived tumors to determine somatic gene mutation(s) defining response to antitumor agents. Taken together, these results describe a novel strategy for developing a pharmacogenomic platform for precision medicine oncology using expandable primary patient-derived tumors.

Authors' Disclosures

M. Kashani-Sabet reports other support from Melanoma Diagnostics, DNARx, LLC; and personal fees from Bristol-Myers Squibb outside the submitted work; in addition, M. Kashani-Sabet has a patent for US-20130344481-A1 issued, a patent for US-9920377-B2 issued, and a patent for US-11142796-B2 issued. No disclosures were reported by the other authors.

Authors' Contributions

S.H. Chang: Conceptualization, data curation, formal analysis, writing—original draft. **R.J. Ice:** Data curation, methodology. **M. Chen:** Methodology. **M. Sidorov:** Methodology. **R.W.L. Woo:** Methodology. **A. Rodriguez-Brottons:** Methodology. **D. Jian:** Methodology. **H.K. Kim:** Methodology. **A. Kim:** Methodology. **D.E. Stone:** Methodology. **A. Nazarian:** Methodology. **A. Oh:** Methodology. **G.J. Tranah:** Formal analysis, funding acquisition. **M. Nosrati:** Methodology, project administration. **D. de Semir:** Data curation, methodology. **A.A. Dar:** Data curation, methodology. **P.Y. Desprez:** Formal analysis, investigation, writing—review and editing. **M. Kashani-Sabet:** Conceptualization, formal analysis, writing—review and editing. **L. Soroceanu:**

Conceptualization, formal analysis, writing–review and editing. **S.D. McAllister:** Conceptualization, formal analysis, supervision, writing–original draft.

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